INITIAL DRAFT REVIEW — completed source-isolated IID comparison; independent learning repeats deferred.

Evidence-guided skill evolution for statutory exemption reasoning

Working initial manuscript, 1 November 2026. Includes the completed chronological learning stream and source-isolated frozen comparison under the source-frozen utility-admission implementation. Extra independent learning builds remain deferred. This manuscript is separate from the preserved v0.23 study. Reference metadata are maintained in the accompanying bibliography.

Abstract

Reusable instructions offer a learning interface for small language models whose weights remain fixed, but an initially useful memory does not establish that further experience improves it. We study this distinction in statutory exemption classification and valuation, where an answer must preserve independently applicable alternatives and assign their protected amounts. Evidence-guided skill evolution combines task-focused execution teaching with complete-output utility admission: a teacher revises legal procedures from released training targets and actual executor errors, while replay-based publication checks both omitted and false claims. A frozen Qwen3-8B learns over 128 sources in eight batches and is then evaluated on 128 source-isolated cases. Its final seven-record library achieves 95.46% classification F1 and 87.90% valuation F1, gains of 30.98 and 32.02 percentage points over the strong empty-library condition. Compared with its own first-batch library, continued evolution adds 11.95 classification points, with a family-adjusted source-paired interval of [7.33, 16.60]. Matched controls connect final-library utility to teacher-visible execution evidence and the admission/selection policy. These results demonstrate continued improvement in legal classification through editable procedures rather than parameter updates.

1. Introduction

Legal rule application requires more than recalling a relevant provision: the answer must connect factual conditions to every applicable alternative and compute its consequence. OpenExempt makes these requirements measurable through generated statutory-reasoning problems with symbolic reference solutions.

External experience offers an editable learning interface for a model whose parameters remain fixed. Natural-language lessons can summarize recurring errors, and evolving playbooks can carry procedures across tasks, as studied by ExpeL and ACE. In a legal application, such a procedure must specify when a provision applies, which alternatives coexist, and how a statutory ceiling becomes an asset-specific amount. These are instructions that the executor must carry out, rather than additional statutory text to recall.

The distinction between a plausible abstraction and a useful procedure matters for experience consolidation. Trace2Skill induces portable procedures from trajectory-local lessons, while ReasoningBank organizes strategies from successful and failed experiences. SkillCommit further evaluates behavioral compatibility before broadening skill scope. Our question concerns the execution of legal alternatives by a small frozen model: can observed claim errors produce procedures that remain useful as new cases arrive? This requires distinguishing the value of acquiring an initial library from the value of continuing to revise it.

We study this question through evidence-guided skill evolution for exemption classification and valuation. After saving a batch's blind forecasts, a teacher receives the released reference outputs, actual parent answers and claim-level errors. It revises one task's instructions while preserving the other task's projection. Edits can change applicability conditions and procedural wording, or add a deidentified execution illustration to an otherwise preserved rule. A shared, bounded textual library is the learned state; the executor continues to generate the legal answers directly.

Publication is tied to the quality of those generated answers. The controller evaluates both the complete proposal and dependency-consistent edit components on all encountered training sources. Admission preserves current-cohort and all-seen native and complete-output F1, while bounding historical mean loss. Complete-output scoring accounts for false claims as well as omissions and incorrect amounts. Retained historical failure witnesses inform later teaching. Together, task-focused revision and utility-based admission connect the content of a legal procedure to its measured execution by the intended model.

Evaluation follows a chronological stream and a subsequent frozen-library comparison on source-isolated cases. A library fixed after the first batch tests the incremental value of later evolution. Same-version controls examine teacher-visible execution evidence and the admission/selection policy, while empty, static and experience-learning libraries provide complementary references. The resulting evidence separates initial knowledge acquisition, subsequent-case learning and final-library utility under a common legal task interface.

The completed comparison gives the frozen executor 95.46% EC F1 and 87.90% EV F1 on 128 new source cases. Relative to its own first-batch library, the final library adds 11.95 EC points; this continuation gain remains positive under the prespecified family adjustment. The contribution is an execution-grounded learning loop for legal procedures, supported by separate evidence for initial library utility, subsequent evolution and the value of teacher-visible execution errors.

2. Related work

Experience-based context adaptation. External learning methods differ in how they represent and consolidate experience. ExpeL extracts natural-language insights from training experiences, and ReasoningBank distills strategies from self-judged successes and failures for later retrieval. ACE maintains an incrementally revised playbook through generation, reflection and curation, whereas Trace2Skill consolidates parallel trajectory-local lessons into a portable skill directory. Our study focuses on procedures for a fixed statutory answer interface, learned from released training labels and observed execution errors, with final-library inference separated from the learning stream.

Behavioral validation and statutory reasoning. SkillCommit preserves instance-specific patches, tests cross-instance transfer and mechanism compatibility, and commits higher-level abstractions that preserve their validated behavior. It also evaluates on OpenExempt, placing legal rule application within behaviorally validated skill evolution. OpenExempt supplies expert-constructed statutory representations, generated cases and computable targets that support analysis of classification and valuation separately. We use that separation to study task-focused procedure revision and admission under both native and complete-output utility. The mechanism controls address how actual execution evidence and publication policy affect learned procedures; the fixed first-batch comparison addresses whether further evolution helps subsequent cases.

3. Task and learning problem

We study whether a frozen language model can improve statutory reasoning by accumulating and revising reusable legal procedures. Each source case supplies facts, statutes and the allowable-jurisdiction result. Exemption classification (EC) identifies every applicable asset–provision alternative. Exemption valuation (EV) additionally assigns the protected amount to each alternative.

The two tasks share source facts but produce independent answers. EV determines eligibility from its own input; the supplied prerequisites identify allowable jurisdictions rather than a solved EC answer.

The target is the benchmark's set of independently applicable alternatives. A specific exemption and a general-property exemption may both be relevant to an asset. Their protected amounts are alternative claims, not an instruction to combine them into an estate-wide allocation. All reference outputs follow the pinned benchmark construction and its supplied statutory text.

Let B denote a textual skill library and F_θ the executor with fixed model parameters. A skill specifies a public legal namespace, supported tasks, an applicability condition, a procedure and exclusions. A shared record can contain separate EC and EV instruction blocks. Given source c and task h, the public namespace and task projection $P_h(B;c)$ select the instructions delivered to the executor:

$$\hat{y}_{c,h}(B) = F_\theta\left(x_{c,h}, P_h(B;c)\right).$$

Here $x_{c,h}$ contains the case facts, supplied prerequisites and statutes. Projection selects relevant instruction text. The executor decides asset eligibility, applicable provisions and claim values. Library tokens, rather than model weights, are the learned state.

The executor uses shared public citation labels and case-local literal asset references. These identify strings in the supplied text. The decoder restores the corresponding strings by lookup. The same interface serves the empty-library and learned-library conditions. EC instructions precede the case; EV instructions follow it. Public asset references carry neither eligibility decisions nor reference amounts.

4. Evidence-guided skill evolution

The learning loop turns observed claim errors into procedures and tests each revision before publication. It has two connected stages: task-focused execution teaching and complete-output utility admission.

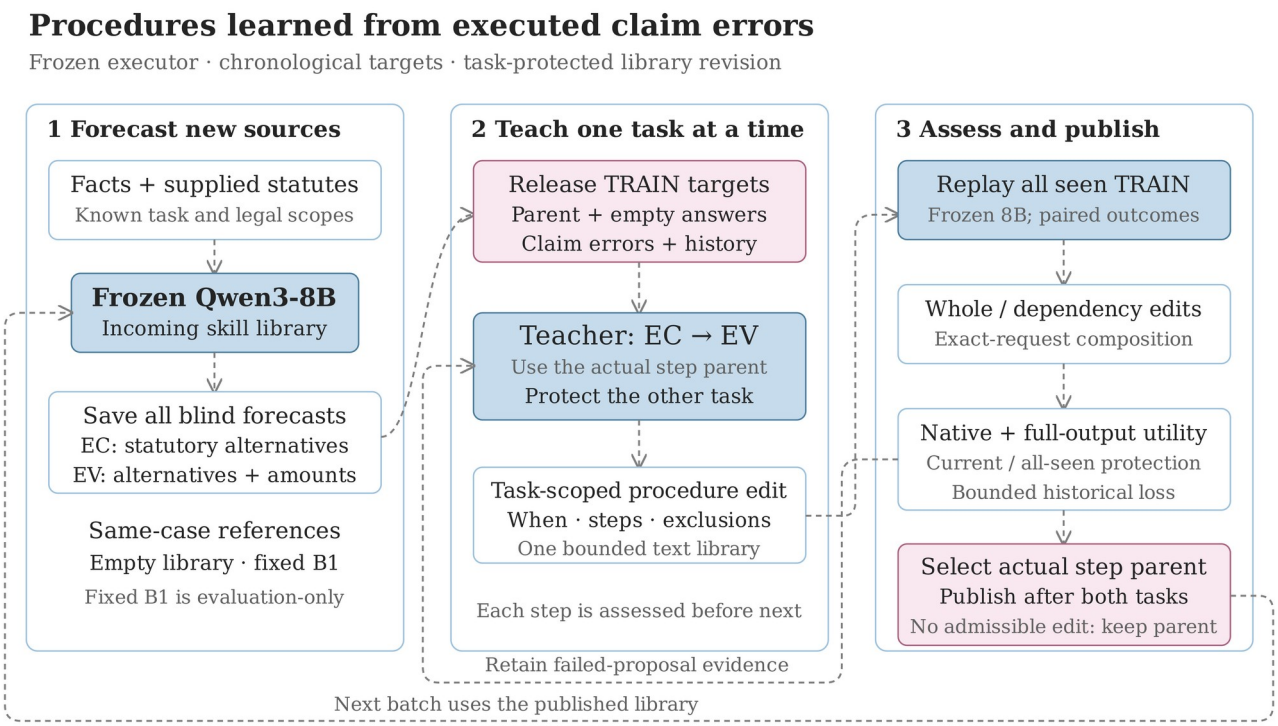


Figure 1. Execution evidence guides task-protected legal procedure revision. All new-source forecasts are saved before training targets are released. The middle and right stages run sequentially for EC and EV: each selected step result becomes the next step's actual parent, and the batch library is published after both tasks. Whole proposals and dependency-consistent compositions are assessed on all encountered training sources under native and complete-output utility. Identical requests reuse the same physical response. Bounded failed-proposal evidence informs subsequent teaching, while the published library serves the next batch. The fixed B1 reference is evaluation-only. At stream completion, the actual final library is frozen for separate evaluation without learning feedback.

4.1. Task-focused execution teaching

Chronological evidence. The learner starts with an empty library and processes successive source batches. It saves all current-batch forecasts before releasing that batch's original reference outputs. Each batch then receives one EC-focused and one EV-focused teaching opportunity. An accepted first-step revision becomes the actual parent of the second step.

The teacher sees the current cases and a bounded historical set. Its evidence includes public facts and statutes, original training targets, actual parent and empty-library answers, and the executor's real response interface. Claim-level feedback distinguishes omitted alternatives, incorrect amounts and extra predictions using the pinned benchmark matcher. Selected model-emitted excerpts supplement these observed outcomes. Original raw outputs remain the evidential source.

Procedures and execution illustrations. The teacher can create, update, merge or remove skill text within the active task projection. It distinguishes an inadequate instruction from an execution lapse. The former can change applicability conditions, procedural wording or exclusions. For the latter, the existing legal prefix is preserved and a compact, deidentified execution illustration is added or replaced. For example, an illustration can show that a general-property alternative must coexist with a qualifying specific alternative, without recording an instance's asset names or values.

Each changed skill–task projection is tied to released training sources. An anchor for an existing rule quotes that same parent projection. A newly introduced projection has no old same-projection

text to quote. Source identifiers and diagnostic rationales remain outside the executor's skill memory.

Task protection. An EC teaching step preserves the EV projection, and conversely. This permits a teacher to improve one procedure without silently rewriting the other task's instructions. Shared records remain possible; their task-specific blocks are checked separately. The complete serialized library, including metadata and both task views, obeys one token budget.

Retained failure evidence. For each teaching task, the next request can retain its latest unsuccessful or partially committed historical proposal, with at most eight actual replay witnesses. A no-op or an unrelated successful edit does not erase that record. Historical candidate outputs retain their original proposed-bank identity. Current-parent reference answers are supplied separately, so an old output is not presented as a fresh execution under the current library.

4.2. Complete-output utility admission

Assessment cohorts. A proposal is assessed on every encountered training source. Let D_t be the current batch, H_t the preceding sources and $A_t = D_t \cup H_t$. For each task, the controller compares a candidate with the actual parent of that teaching step. Bounded teacher communication therefore does not reduce the assessment cohort.

Let $g_{S,h}(B)$ be the mean score of library B on source set S and task h. We use both the original native F1 and complete-output F1 in the admission constraints:

$$g_{D_t,h}(B') \geq g_{D_t,h}(B), g_{A_t,h}(B') \geq g_{A_t,h}(B),$$

$$g_{H_t,h}(B') \geq g_{H_t,h}(B) - \varepsilon.$$

The historical constraint applies when H_t is nonempty. Every inequality is checked separately for EC and EV, and for both score definitions. An admissible change must also strictly improve a current-cohort native or complete-output score, or reduce a current-cohort format-failure or negative/unscored-asset claim count. Invalid responses retain zero scores. False claims remain penalized by complete-output F1. Format and false-claim counts are reported as diagnostics rather than additional zero-increase vetoes.

These constraints preserve current and all-seen aggregate quality while allowing bounded historical trade-offs. They evaluate the realized training executions; individual cases can improve or worsen. The token budget and task-protection constraints apply to the same delivered candidate.

Whole and dependency-consistent revisions. The controller considers the complete proposal and a composition of admissible legal edit components. Components follow actual co-visible changed instructions in public legal contexts. Their number depends on the proposal and encountered contexts. A composition retains parent instructions outside the admitted components and publishes one fixed library.

Every composed request must match a previously measured parent or proposed request in its complete messages, schema and generation profile. The existing raw response is then interpreted under the actual composed library. This yields library-level selection without constructing a case-specific mixture of favorable answers.

Selection and publication. Among admissible whole and composed revisions, selection maximizes the changed task's all-seen complete-output gain:

$$G_h(B', B) = g^{full}_{A_t,h}(B') - g^{full}_{A_t,h}(B).$$

An exact tie retains the whole proposal. The implementation uses the minimum and then mean task-wise gains; under task protection the unchanged task contributes zero, making this ordering equivalent to G_h. If neither revision is admissible, the parent remains. After both teaching steps, the actual publication is checked against the original batch parent with the same admission contract.

The loop freezes its actual last-batch library for evaluation. The model generates direct EC answers and numeric EV claims throughout; numerical operators are excluded from skill records. Exact-request reuse shares identical responses and charges their generation once. It does not turn aliases into independent samples.

5. Experimental design

We address four research questions. **RQ1:** How useful is the final frozen library relative to empty, static and experience-learning libraries on new sources? **RQ2:** Do later revisions improve subsequent-case predictions beyond the library acquired after B1? **RQ3:** What does teacher-visible execution evidence add when training labels and replay-based admission remain available? **RQ4:** How does complete-output utility admission compare with the strict admission/selection policy? The frozen comparison addresses RQ1, the chronological and final-library B1 comparisons address RQ2, and the two same-version learning controls address RQ3 and RQ4.

5.1. Data and chronological learning

The training stream contains 128 source cases generated with the pinned OpenExempt procedure, with one EC and one EV view per source. We retain the original facts, reference outputs, order and statutory material. Eight batches contain 16 sources each. Current-batch targets are released after all forecast conditions have been saved. There is no validation split.

The source-isolated IID confirmation contains 128 additional source cases produced by the same generator and construction settings. Its construction excludes source keys from the training and previously used packets. Confirmation inputs were opened after the method, required controls, actual final-bank hashes, common inference conditions and statistical family had been bound. Previously inspected evaluations remain separate from this confirmation.

5.2. Comparators and mechanism controls

The comparison includes a strong empty-library executor, a static law-only library, retained experience-learning libraries and the learned final library. The same ACE library is evaluated through its native interface and a lossless common-interface wrapper. These are two interfaces to one learned bank. Trace2Skill–Text supplies an offline, text-only trajectory-learning comparison. Both adaptations use the released training references and the same requested teacher model and reasoning effort; Appendix C specifies their formation conditions. Their existing completed libraries are reused; new test predictions follow the shared source matrix.

Two same-version controls address the current mechanisms. The strict-policy condition retains the teacher, anchors and bounded history while restoring the original zero-regression, count-veto and native-gain selection policy. This tests the admission/selection policy bundle. The teacher-evidence condition retains the utility policy, labels, facts, statutes, current parent and public executor interface while withholding explicit actor outcomes and outcome-labelled feedback from the teacher. Its controller still performs actual replay and admission, so the condition measures the incremental value of teacher-visible execution evidence.

A completed predecessor library is retained as a historical method-version reference, separate from matched component controls. Extra independent main-method, ACE and Text learning builds are deferred in the initial study. Both control builds completed all eight batches. The frozen comparison binds all ten actual libraries before prediction.

5.3. Metrics and learning curves

Primary endpoints are the original native per-case EC and EV F1, averaged separately across sources. The pinned evaluator's asset matching and amount tolerance remain unchanged. Complete-output F1 pools true positives, false positives and false negatives within each source and then averages across sources. It includes claims on unmatched or negative-target assets. Consequently, native and complete-output F1 use different within-case aggregation and need not have a fixed numerical ordering.

The chronological curve reports first answers on each new batch before its targets are released. We compare the incoming library with the same-case empty library and a snapshot fixed after exactly the first batch. This snapshot is an evaluation condition, not a teacher or admission input. B2 starts from that same library; subsequent batches measure the effect of further evolution. Post-label assessment replays are reported separately.

Evaluation saves every condition's predictions before unified scoring. Source-paired resampling keeps each case's EC and EV views together and quantifies uncertainty conditional on the actual frozen libraries. The contrast family is fixed before confirmation inputs are opened. Reports include invalid outputs, omissions, incorrect amounts, extra claims, actual skill edits and delivered task blocks. Physical native origins, exact-request reuse, teacher responses and attempts, token totals and elapsed time provide cost accounting.

5.4. Implementation settings

Setting	Value
Executor	Qwen3-8B, fixed weights, BF16
Runtime	One A40 model instance; four native workers; one teacher worker
Inference software	vLLM 0.10.2; PyTorch 2.8.0
Teacher request model	`gpt-5.6-sol`, high reasoning effort
Training sources / batches	128 / 8
Sources per batch	16
Teaching opportunities	One EC step and one EV step per batch; at most 16 successful responses
Complete-library budget	4,096 executor-tokenizer tokens
Historical mean-loss allowance ε	0.01 on the F1 scale, or one percentage point
Executor context / maximum output	32,768 / 8,192 tokens
Decoding	Seed 45; temperature 0.6; top-p 0.95; top-k 20; thinking enabled
Teacher historical proposal memory	At most one per task; eight witnesses per proposal
Upstream generator/evaluator revision	`ec6fa2547431096588ce2d3ef107b8adc33cd4fe`

6. Results

6.1. Final-library utility on new sources (RQ1)

The final library improves both statutory classification and valuation over the empty, static and retained experience-learning libraries (Table 1). It reaches 95.46% EC F1 and 87.90% EV F1, compared with 64.48% and 55.88% for the strong empty executor: gains of 30.98 and 32.02

percentage points. Relative to the strongest retained experience condition, ACE with the common interface, the gains are 28.87 and 20.98 points. All four corresponding family-adjusted intervals are positive (Appendix B.4). The fixed backbone keeps model capacity unchanged, and the common-interface comparison shows that changing ACE's delivery interface does not account for the learned library's advantage.

Table 1. Frozen-library performance on 128 source-isolated IID cases (%). EC and EV are the original source-mean native F1 endpoints; Avg. is their descriptive mean. Bold denotes the highest point estimate in each column. All conditions retain invalid and unfavorable outputs; formation conditions and costs are reported separately.

Frozen library	EC F1	EV F1	Avg.
Strong empty	64.48	55.88	60.18
Law-only static	60.40	55.34	57.87
ACE native interface	65.00	58.68	61.84
ACE common interface	66.59	66.92	66.75
Trace2Skill–Text	55.08	53.31	54.19
Own frozen B1	83.51	84.25	83.88
Strict-policy control	80.29	**89.01**	84.65
Execution-evidence-withheld control	86.98	80.38	83.68
Predecessor library	87.63	83.91	85.77
Evidence-guided evolution	**95.46**	87.90	**91.68**

The comparison comprises 2,560 actual predictions, all saved before unified scoring. ACE's two rows use one learned playbook through two interfaces. The predecessor is a method-version reference, while the strict-policy and teacher-evidence conditions are matched learning controls. Table B3 additionally reports complete-output quality, exact-target matching and format failures.

6.2. Chronological evolution (RQ2)

Continued evolution improves new-case answers beyond the first learned library. Across all 112 sources in B2–B8, the evolving library raises native EC F1 by 10.96 percentage points and EV F1 by 3.42 points over its own fixed B1 library (Table 2). Complete-output gains are 10.27 and 3.44 points. The two libraries coincide at B2, so their separation from B3 onward measures further learning after initial library construction rather than the benefit of receiving any skills.

Table 2. First-answer performance on the 112 new training-stream sources after B1 (%). Each source contributes one EC and one EV answer. All B2–B8 forecasts are included; the last row is the paired evolving-minus-fixed-B1 mean in percentage points.

Library at first encounter	EC native F1	EV native F1	EC complete-output F1	EV complete-output F1
Empty	63.21	61.43	66.56	61.29
Fixed after B1	80.87	84.74	84.15	83.97
Evolving	91.83	88.16	94.42	87.41
Evolving − fixed B1, pp	+10.96	+3.42	+10.27	+3.44

Actual requests establish where later revisions reach the executor. Of the 112 subsequent EC tasks, 86 receive guidance different from fixed B1; native F1 improves on 52 and decreases on three. For EV, 96 tasks receive changed guidance, with 36 improvements and 14 decreases. The remaining 26 EC and 16 EV tasks have identical inputs and reuse the same recorded responses. Same-case references separate library utility from changes across batches. For example, all three EV conditions decrease at B7, while the evolving library remains 2.25 points above fixed B1; B8's paired advantage is 6.44 points. Figure 2 retains these observed fluctuations and reports both scoring views of every batch.

The library evolves mainly by revising existing procedures. Nine of the 16 teaching steps commit an update: five EC steps and four EV steps. After B1, the accepted edits make 16 revisions to existing task projections and add one EV projection while retaining seven records. The final library occupies 3,993 tokens, and each subsequent first-answer task receives two or three projected instruction blocks (Appendix D). These records distinguish content evolution from growth in the number of entries.

The final-library comparison tests whether this evolution also benefits a separate source set. On the 128 confirmation cases, final versus own-B1 EC F1 increases from 83.51% to 95.46%, a paired gain of 11.95 points with family-adjusted interval [7.33, 16.60]. EV increases from 84.25% to 87.90%, a 3.65-point estimate with adjusted interval [-2.82, 10.28]. The classification continuation gain therefore persists beyond the learning stream; the valuation estimate does not establish a corresponding adjusted-significant continuation effect.

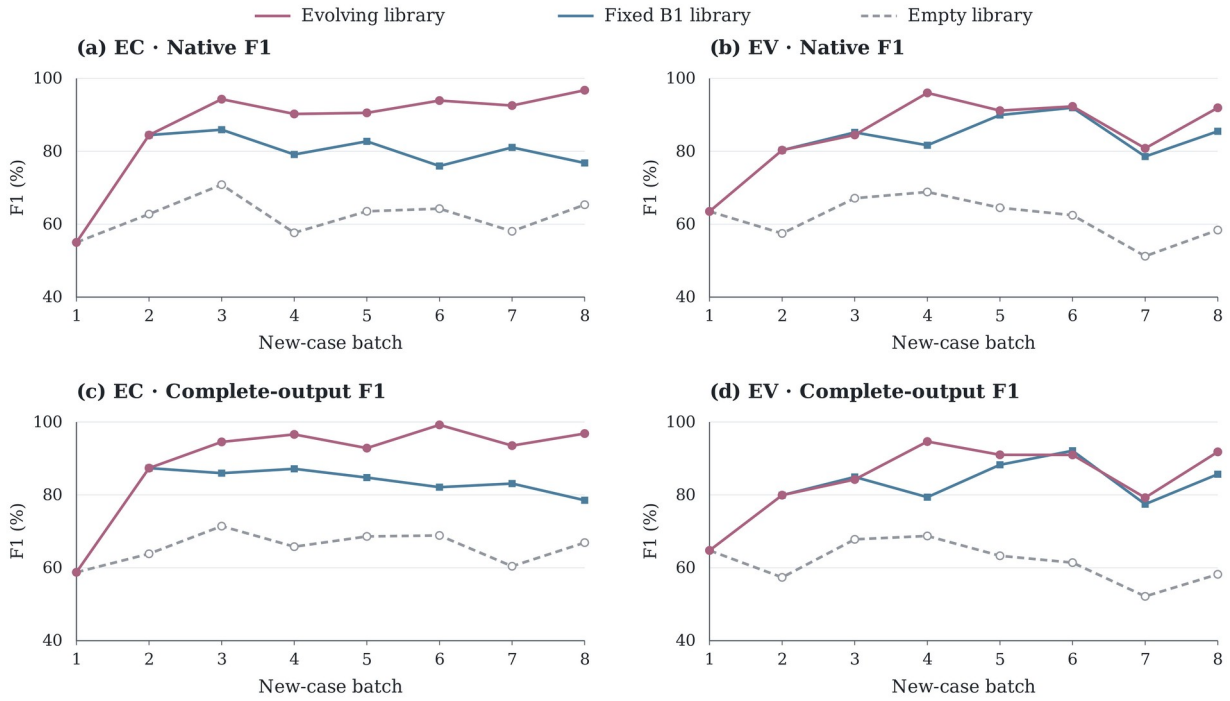


Figure 2. Continued evolution improves subsequent-case performance beyond the first learned library. Each point is the mean over 16 new source cases, whose EC and EV first answers are saved before the current targets are released. Panels show native and complete-output F1 for the evolving, empty and fixed first-batch libraries. The fixed B1 library becomes available at B2; B1 starts empty. All eight observed batches are shown without smoothing. Post-label replay scores are excluded.

6.3. Execution evidence and publication policy (RQ3, RQ4)

Teacher-visible execution evidence (RQ3). Hiding explicit executor outcomes from the teacher while retaining facts, statutes, released targets and replay-based admission yields final EC/EV F1 of

86.98/80.38%. The complete method improves these endpoints by 8.48 points [3.91, 13.59] and 7.53 points [2.38, 12.67], respectively, where brackets give family-adjusted source-paired intervals. This supports grounding revisions in observed execution errors rather than reference solutions alone under the retained learning protocol. The control still receives labels and public solved material, and its controller supplies indirect feedback through replay and admission.

Chronological evidence distinguishes this effect from initial acquisition. On the same 112 post-B1 sources, the evidence-withheld library records native EC/EV F1 of 84.54/81.27%, compared with 84.54/81.91% for its own B1 snapshot. Its continuation changes are 0.00/-0.64 points, whereas the main method changes by +10.96/+3.42 points (Table 2). Appendix D.5 reports both complete controls and their learning-path boundaries.

Admission and selection policy (RQ4). Complete-output utility admission gives a 15.17-point EC advantage over the strict-policy control, with adjusted interval [10.53, 20.35]. The strict control instead has the highest EV point estimate, 89.01% versus 87.90%; the paired utility-minus-strict EV difference is -1.11 points [-6.20, 3.65]. The supported policy advantage is therefore classification quality rather than superiority on every endpoint. Because the control changes admission and selection together, this comparison evaluates the policy bundle and its resulting learning path, not an isolated count-veto effect.

Complete-answer quality. The main library attains 96.67% complete-output EC F1 and 87.51% complete-output EV F1, with exact-target matching of 57.81% and 41.41% (Table B3). These measures distinguish high claim coverage from an entirely matched answer. Its two format failures are retained as zero-score EV outcomes. Complete-output evaluation also retains extra claims on unmatched or negative-target assets, so a native-F1 improvement does not remove their cost from the accompanying quality assessment.

7. Discussion

Execution-grounded legal procedures. The completed source-isolated evaluation shows that an editable library can substantially improve a small frozen model’s statutory answers. The teacher-evidence control connects this utility to information about the model’s actual omissions, incorrect amounts and extra predictions. The teacher acts during learning; the frozen executor still selects provisions and amounts itself at test time. The learned state therefore consists of reusable instructions rather than a second answer-generating model or a privileged numerical solver.

Improvement after initial acquisition. A useful initial library and a useful evolution process answer different questions. The own-B1 reference holds initial knowledge fixed while later procedures are revised. Its paired comparison supports further classification improvement both on subsequent learning-stream sources and on the separate frozen evaluation. The positive EV continuation estimate remains uncertain under family adjustment. Distinguishing these effects makes continued learning testable without assuming monotonic improvements across changing batches.

Evolution within bounded memory. The stream retains seven records while revising 16 existing task projections after B1 and adding one EV projection. Actual delivery traces and the own-B1 contrast connect these changes to the instructions used on later cases. Task protection permits targeted revision inside shared records, while complete-output assessment charges both omissions and false claims. The resulting memory objective concerns the utility of applicable procedure text, not growth in the number of entries.

Interpretation boundary. The evidence concerns statutory alternatives under supplied facts, statutes and jurisdiction prerequisites in the pinned generator. The strict-policy comparison tests a bundle of admission and selection choices; it does not show a uniform valuation advantage. Source-

paired uncertainty describes these retained learned libraries, with independent learning builds reported separately when available.

8. Conclusion

Task-focused execution teaching and complete-output utility admission turn observed legal errors into reusable skills for a frozen 8B model. Across 128 new OpenExempt sources, the final library attains 95.46% EC and 87.90% EV F1 and adds 11.95 EC points beyond its first-batch library.

Appendix A. Learning algorithm and evidence routing

A.1. Chronological library updates

The executor parameters, task interface, decoding profile and source order remain fixed throughout a build. The initial library is empty. The following sequence implements one batch; its output library becomes the next batch's incoming library.

1. **Forecast.** Generate and save both task views of every current source under the incoming library, the empty library and the fixed first-batch library. The first-batch reference is empty during B1 and is set immediately after B1 publication. Two retained reference aliases, the frozen initial library and direct-numeric empty condition, also pass through the same request path. In this build the initial library is empty and every condition uses direct numeric claims. Identical requests share one physical response.
2. **Release.** Read current-source targets after all forecast conditions have been saved. Save original forecast scores separately from the subsequent teaching and replay records.
3. **Teach EC.** Assemble the actual parent library, current-source evidence, selected historical evidence and retained EC proposal feedback. Request one diagnosed edit. Check the edit against its released evidence, legal anchors, protected EV projection and complete-library token budget.
4. **Assess and select.** A changed, valid proposal is executed on all encountered sources. Check the whole proposal and dependency-consistent components under Section 4.2. If a composed library is admissible, verify the complete request identity for its reused executions. Compare admissible whole and composed libraries, publish the selected change as the next step's parent, or retain the existing parent.
5. **Teach EV.** Repeat the same process for EV using the actual result of the EC step. Protect EC instructions. The two opportunities concern different tasks, rather than repeated attempts on whichever task is easier to improve.
6. **Publish the batch.** Check the resulting library against the original batch parent. Save the actual library, both teaching decisions, assessment records and parent-to-publication lineage. After exactly B1, copy the actual publication into the fixed first-batch reference; never replace that reference using later performance.

A no-op, an invalid edit or an over-budget proposal consumes its scheduled teaching opportunity without an extra same-task retry. The completed stream returns its actual B8 publication. A predeclared forward-harm stop examines two consecutive batches from B2 onward: mean native EC/EV loss greater than two percentage points against a retained empty/initial/direct reference, or EV complete-output loss greater than two points against the initial reference. Such a stop terminates the build rather than selecting a favorable earlier checkpoint. The first-batch frozen reference supplies evaluation evidence and is not used in teaching or admission.

A.2. Communication and execution evidence

Each task-focused request includes every current source and selects up to 16 historical background sources. Mandatory retained failure witnesses are included first, followed by public-context

witnesses, observed regressions and low-scoring parent cases, with source identifiers breaking ties. If mandatory witnesses exceed the background limit, the limit expands to include them. This is communication selection within already encountered training data. Proposal assessment still covers every encountered source.

The supplied public law corpus is deduplicated within a request. Current-parent answers contain the actual parsed output, native and complete-output scores, and claim-level errors. For each selected parent execution, the teacher can also see the final native submission and up to the last 1,024 characters of model-emitted reasoning. A malformed final submission contributes bounded head and tail excerpts. These excerpts are identified as unverified model emissions; they do not replace the original output or reference target.

Retained proposal feedback identifies the historical candidate and its original parent, while references to current-parent answers point to separately supplied observations. Each task retains at most one unsuccessful or partially committed proposal with at most eight replay witnesses. The controller distinguishes a fully committed proposal, a partially committed proposal and a rejected proposal. The executor receives the resulting procedures, not the source identifiers, diagnostic labels or historical answer records.

A.3. Exact-request accounting

A request identity contains the full messages, response schema and generation profile. A library hash alone is insufficient: different libraries can deliver the same task projection, while changed applicable text produces a different request. Every reused response retains its original raw-call identity and is decoded for the actual requesting condition. A dependency composition is valid only when each composed request matches a measured parent or proposal request. Consequently, composition selects a fixed library without an additional generation round or per-case score-based answer choice.

A fixed sampling seed specifies a decoding condition, not guaranteed bitwise repeatability of fresh online calls (version-pinned serving documentation). Exact-request reuse shares the original recorded response rather than issuing a fresh repetition.

Cost reporting counts each physical native origin once and lists logical forecasts, assessment views and reuse separately. Teacher responses and transport attempts are distinct quantities. Provider-reported cached input is a subset of input tokens, and reasoning tokens are a subset of output tokens; neither is added a second time. Actual total tokens and elapsed times, rather than assumed equal compute, describe each learning method.

Appendix B. Prespecified frozen comparison and uncertainty

B.1. Source and library boundaries

The confirmation recipe fixes 128 source cases with seed 2026103001. It excludes the source keys of 564 cases across 15 earlier packets. Each source contributes one EC and one EV question. This is source-isolated IID evaluation within the pinned generator, rather than a claim of unseen-jurisdiction or official Advanced-benchmark evaluation.

The prespecified matrix contains ten conditions: empty, static, ACE native, ACE common-interface, Text, the evolved B8 library, its own fixed B1 library, the strict-policy control, the teacher-execution-evidence-withheld control and the completed predecessor library. It therefore contains 2,560 logical predictions. The two ACE conditions share one learned bank; the predecessor is a version reference, not a matched component ablation.

Before opening confirmation questions, the evaluation binds each real learned bank to its configuration, frozen source manifest, completion, publication and terminal preservation. The B1

reference additionally binds its actual first-batch snapshot. Learning-source snapshots establish the executor and grader shared across teaching versions. All condition forecasts are saved before confirmation targets are read for unified scoring. Evaluation performs no teacher update or skill revision.

B.2. Scores and paired contrasts

For a condition and task, the primary mean is the arithmetic mean of its 128 original per-case native F1 scores. Complete-output scoring additionally matches all target assets and counts predictions on unmatched or negative-target assets. It pools claim-level true positives, false positives and false negatives within each case, then averages case scores. EC deduplicates predicted citation strings; EV uses the pinned claim matcher and amount tolerance. Invalid outputs retain zero, and zero-denominator conventions follow the pinned scorer.

For a fixed comparator, source i supplies the paired difference between the evolved library and that comparator. The reported effect is the mean of these 128 differences, with EC and EV reported separately. Source-paired bootstrap draws resample whole source indices, preserving both tasks and all conditions together. The procedure uses 20,000 draws from NumPy PCG64 with seed 2026103092 and linear quantiles.

The evolved library is compared with the other nine conditions on both native endpoints. These 18 contrasts form one primary family. Reports provide ordinary 95% intervals and Bonferroni-adjusted intervals with quantiles $0.025/18$ and $1-0.025/18$. Complete-output quality, exact-target scores, error categories and cost are descriptive accompanying evidence. The bootstrap characterizes source variation conditional on the actual learned libraries; it is not a substitute for independent learning builds.

B.3. Complete-output quality and invalid responses

Table B3. Accompanying quality endpoints on all 128 cases per task (%). Paired values are EC / EV. Exact-target matching compares the decoded full answer with the reference; format-failure counts use the original parser. Complete-output F1 is the per-case pooled-claim score averaged across sources, not a global ratio over all claims.

Frozen library	Complete-output F1 EC / EV	Exact target EC / EV	Invalid EC / EV
Strong empty	66.65 / 56.51	11.72 / 8.59	0 / 1
Law-only static	63.64 / 56.37	9.38 / 8.59	0 / 3
ACE native interface	66.86 / 59.51	14.06 / 10.94	6 / 4
ACE common interface	69.73 / 66.41	17.19 / 14.84	1 / 1
Trace2Skill–Text	58.88 / 54.78	11.72 / 7.81	5 / 1
Own frozen B1	85.25 / 83.75	27.34 / 35.16	0 / 2
Strict-policy control	81.60 / 88.28	26.56 / 44.53	2 / 1
Execution-evidence-withheld control	88.51 / 79.84	44.53 / 26.56	0 / 0
Predecessor library	88.86 / 83.33	32.81 / 30.47	0 / 1

Frozen library	Complete-output F1 EC / EV	Exact target EC / EV	Invalid EC / EV
Evidence-guided evolution	96.67 / 87.51	57.81 / 41.41	0 / 2

All 30 invalid outputs correspond to original output-length terminations and retain zero scores. The main library has two such EV outcomes. The original grader does not emit complete pooled counts for those malformed outputs, so no EV global micro-average is reconstructed from a favorable subset or repaired answer.

B.4. Prespecified paired effects

Table B4. Evolved-library minus comparator native F1 (percentage points). Each cell gives the paired effect and its family-adjusted interval. All 18 EC/EV endpoints share the prespecified Bonferroni family; 15 adjusted lower bounds are positive. Ordinary 95% intervals, case-level differences and source keys remain in the machine-readable results.

Comparator	EC difference [adjusted interval]	EV difference [adjusted interval]
Strong empty	+30.98 [24.80, 37.51]	+32.02 [23.79, 40.39]
Law-only static	+35.06 [28.48, 41.91]	+32.57 [24.69, 40.87]
ACE native interface	+30.46 [24.05, 37.49]	+29.22 [21.67, 37.30]
ACE common interface	+28.87 [22.97, 34.79]	+20.98 [14.63, 27.43]
Trace2Skill–Text	+40.39 [32.92, 48.30]	+34.59 [26.96, 42.55]
Own frozen B1	+11.95 [7.33, 16.60]	+3.65 [-2.82, 10.28]
Strict-policy control	+15.17 [10.53, 20.35]	-1.11 [-6.20, 3.65]
Execution-evidence-withheld control	+8.48 [3.91, 13.59]	+7.53 [2.38, 12.67]
Predecessor library	+7.83 [4.57, 11.16]	+3.99 [-1.66, 9.86]

The EV contrasts with own B1, strict policy and the predecessor have adjusted intervals spanning zero. No equivalence or non-inferiority conclusion is inferred from these intervals. Tables 1, B3 and B4 are generated from the unchanged completed-run metrics and paired statistics in `paper/tables/utility_frozen_results_20261101.json`.

Appendix C. Experience-library formation conditions

The retained ACE and Trace2Skill–Text adaptations use the same 128 training sources and original EC/EV references. Their configurations match the main method's Qwen3-8B decoding profile and request `gpt-5.6-sol` with high reasoning effort. ACE receives the reference and actual prediction at every used reflection; Trace2Skill–Text analyzes each of the 256 empty-library trajectories with its reference, score and actual output. These are task-interface adaptations, rather than reproductions of the authors' agentic environments.

Learning condition	Library formation	Complete-library token cap	Teacher responses
Execution-evidence evolution	Online, task-focused revision and utility admission	4,096	16 used
ACE adaptation	Online reflection, regeneration and incremental curation	8,192	423 used in the retained stream

Learning condition	Library formation	Complete-library token cap	Teacher responses
Trace2Skill–Text	Offline trajectory analysis, hierarchical merge and text translation	8,192	321 used

ACE's retained stream contains 415 reflections and eight batch curations. Its continuation records separately preserve 19 successful but uncommitted teacher responses; failed-request usage is unknown rather than zero. Trace2Skill–Text uses 256 trajectory analyses, 64 merges and one translation, followed by a training replay that does not select its final bank. Historical formation schedules and budgets remain distinct from the main method's four-worker learning schedule. The final comparison standardizes executor inference over these frozen libraries; formation costs are reported separately.

Appendix D. Observed skill evolution and compute

D.1. Published library changes

Table D1 reports actual committed task projections rather than proposed operations. A shared record can hold separate EC and EV blocks, so a new task projection need not add a record. Each batch has one teaching opportunity per task.

Batch	Records	EC / EV blocks	Serialized tokens	Committed EC projection edits	Committed EV projection edits
B1	7	7 / 6	2,461	7 new	6 new
B2	7	7 / 6	3,257	4 revised	5 revised
B3	7	7 / 6	3,413	3 revised	0
B4	7	7 / 6	3,413	0	0
B5	7	7 / 6	3,594	1 revised	0
B6	7	7 / 6	3,654	2 revised	0
B7	7	7 / 7	3,975	0	1 new
B8	7	7 / 7	3,993	0	1 revised

Each post-B1 first-answer task receives two or three blocks selected by its public task and supplied legal namespaces. Delivery records establish which instructions the executor received; answer scores measure whether those instructions helped. Forecasts use the incoming published library, so edits made after B8's first answers contribute to the final-library comparison rather than retrospectively to B8's curve.

The evidence-withheld control finishes with 12 records occupying 3,965 tokens. It retains two revisions after B6 and one after B8. The B6 revisions reach the B7–B8 forecasts; the B8 revision first reaches new cases in the frozen-library comparison.

D.2. Actual build cost

The totals cover this complete learning build, including its forecast reference conditions and training-assessment replays. Each physical native origin is counted once; exact-request reuse retains that recorded response.

Quantity	Recorded value
Physical native generations	1,207
Exact-request reuses	3,497
Native input / output tokens	8,293,891 / 3,196,466
Successful teacher responses / recorded attempts	16 / 16
Teacher input / output tokens	1,778,335 / 66,439
Cached input / reasoning output subsets	115,968 / 38,939
Service wall time	9.00 h

Cached input is included in teacher input, and reasoning output is included in teacher output. Service wall time is measured from the manager's monotonic start and success events; it includes teacher waits, CPU work and terminal preservation, and is not a measurement of active GPU hours. Frozen-test inference and other learning builds are outside these totals.

D.3. Teacher and executor work across retained builds

The main build uses fewer teacher tokens and more local generations than the two retained adaptations (Table D3). Local counts include each build's recorded forecast references and training-assessment replays. Teacher and executor tokens are reported separately because the two roles use different models; their counts do not imply a common GPU-time or monetary cost.

Table D3. Recorded response generation in the actual retained learning streams. Teacher input includes cached input, and teacher output includes reasoning output. Formation schedules and memory budgets are specified in Appendix C.

Learning condition	Role	Responses	Input tokens	Output tokens
Execution-evidence evolution	Teacher	16	1,778,335	66,439
Execution-evidence evolution	8B executor	1,207	8,293,891	3,196,466
Strict-policy control	Teacher	16	1,776,813	67,501
Strict-policy control	8B executor	1,206	8,172,956	3,466,541
Teacher-execution-withheld control	Teacher	16	904,633	46,046
Teacher-execution-withheld control	8B executor	974	6,587,209	2,416,560
ACE adaptation	Teacher	423	11,526,040	405,699

Learning condition	Role	Responses	Input tokens	Output tokens
ACE adaptation	8B executor	896	7,122,372	2,258,447
Trace2Skill–Text	Teacher	321	6,429,967	503,252
Trace2Skill–Text	8B executor	512	4,061,370	1,463,066

ACE's operational continuations additionally retired 19 successful teacher responses, containing 458,657 input and 21,354 output tokens, and 15 executor generations, containing 135,867 input and 45,681 output tokens. These are preserved separately from the retained-stream totals. Failed-request usage that was not reported remains unknown rather than zero.

D.4. Completed frozen-evaluation cost

The ten-condition frozen comparison uses 2,560 physical generations and no prior-stage answers or exact-request aliases. Total executor input/output is 19,242,658/6,841,407 tokens; teacher calls are zero. The original completion occurs after 19 h 50 min 03 s of wall time. This covers all conditions jointly and is not an estimate of active GPU hours or a main-method-only inference cost. The original 2,530 stop and 30 length outcomes are retained. All forecasts precede unified target access and scoring.

D.5. Learning-path evidence for the mechanism controls

Table D5. Chronological first answers on the same 112 post-B1 sources (%). These are learning-stream forecasts, not confirmation scores or post-label replay.

Learning condition	EC native F1	EV native F1	EC complete-output F1	EV complete-output F1
Evidence-guided evolution	91.83	88.16	94.42	87.41
Strict policy	76.31	83.49	78.70	83.04
Execution evidence withheld	84.54	81.27	87.15	80.36

The strict-policy B1 library has no EV projection, so its later EV gain relative to that snapshot also includes first acquisition; it is not a matched post-acquisition effect. An audit holding the main run's recorded outputs fixed finds that all nine actual commits also satisfy the strict gate. Consequently, the complete-build comparison cannot attribute its difference solely to a count veto: the policy and induced teaching path are jointly changed.

The evidence-withheld condition keeps released targets and public solved material. Its fixed-B1 native EC/EV F1 is 84.54/81.91%, and its complete-output continuation changes are 0.00/-0.72 points, compared with +10.27/+3.44 for the main method. Hiding explicit teacher-visible execution evidence does not remove controller feedback or establish that labels alone cannot produce useful skills.

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